Solvency Field Theory

A Reader's Guide to the V7 Framework

From Information Thermodynamics to Dark Energy, Galaxy Dynamics, and Exact General Relativity

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Abstract

Solvency Field Theory (SFT) proposes that cosmic expansion is the thermodynamic cost of irreversible information processing. The framework rests on a single master equation—$H = \dot{S}/I$—identifying the Hubble rate as the fractional rate of entropy accumulation relative to the universe's stored information. This document serves as a concise guide to the V7 framework, presenting the core claim, three principal results spanning cosmological, galactic, and strong-field regimes, and a map to the modular proof documents in the V6.1 repository. No new derivations are introduced here; instead, this note provides the narrative lens through which the full evidence bundle should be read.

1. The Core Claim

Three established principles constrain the geometry of an information-processing universe:

Landauer's Principle. Erasing one bit of information dissipates at least $k_B T \ln 2$ of energy as heat. Information processing has irreducible thermodynamic cost.

The Holographic Bound. The maximum information content of a spatial region scales with its boundary area, not its volume (Bekenstein 1973, 't Hooft 1993, Susskind 1995). Space has finite information capacity.

Jacobson's Thermodynamic Identity. Einstein's field equations can be derived as an equation of state for horizon entropy (Jacobson 1995). Gravity is thermodynamic.

If the universe continuously generates entropy (second law), and no spatial region can exceed its holographic information density (holographic bound), then geometry must expand to accommodate new information. The expansion rate must equal the fractional rate of information accumulation:

$$\mathbf{H = \dot{S}_{total} \ / \ I}$$

where $\dot{S}_{total}$ is the total entropy production rate and I is the accessible stored information (radiation entropy of the CMB and cosmic neutrino background). This is not postulated as a new law—it is the unique fractional rate consistent with all three principles. It is a thermodynamic identity, analogous to $T = 2E/(3Nk_B)$ in statistical mechanics: a recognition of what expansion *means* in an information-theoretic universe.

2. Three Regimes, One Equation

The master equation governs three distinct observational regimes. In each, it produces quantitative results that can be compared to observation:

Regime	Governing Variable	Scale	Key Result
Cosmological	$\dot{S}$ decomposition into active + passive channels	Gpc	$w_0 = -0.837$, $w_a = -0.615$
Galactic (weak-field)	g/a_SFT where a_SFT = $cH_0/2\pi$	kpc	63% error reduction on 137 SPARC galaxies
Strong-field	$\chi = GM/2c^2r$ (compactness burden)	km	Exact Schwarzschild metric, exact TOV

These are not three separate theories. They are three consequences of the same master equation operating at different scales. The cosmological sector governs the global expansion history. The galactic sector emerges when local gravitational accelerations drop below the cosmic processing rate. The strong-field sector governs compact objects where the compactness burden approaches unity. Each regime flows from $H = \dot{S}/I$ and its local consequences.

3. The Cosmological Sector: Dark Energy from Structure Formation

The dark energy equation of state w(z) follows from differentiating the master equation. The entropy production rate $\dot{S}_{total}$ decomposes into two physical channels, computed directly from IllustrisTNG TNG300 simulation data (~700,000 supermassive black holes across z = 0 to z = 2):

The Active Channel (Baryonic Processing). Bekenstein-Hawking entropy from black hole accretion ($\Sigma M_i \cdot \dot{M}_i$) plus stellar entropy from star formation. This channel peaks at $z \approx 1$–2 and crashes by ~80% to the present epoch. It drives the evolutionary running w_a.

The Passive Channel (Halo Architecture). The holographic occupancy cost of maintaining virialized dark matter halos (M_{200c}). Dark matter does not radiate or accrete, but its organized gravitational architecture demands ledger space on the holographic boundary. This channel sets the present-day depth w_0.

Result: $w_0 = -0.837$, $w_a = -0.615$. The Pantheon+ best-fit is $w_0 = -0.841$, $w_a = -0.604$. Distance in the CPL plane: **0.005**. No cosmological constant. No new scalar fields.

Three Proof Tests

Ablation. Removing the passive halo channel collapses the DESI posterior match by 36–1263 log-units. The channel is load-bearing, not cosmetic.

Structural Specificity. Virialized halo mass (M_{200c}) works: $w_0 = -0.835$. Raw dark matter mass fails completely: $w_0 = -1.065$, $w_a = +0.080$. The holographic ledger taxes organized gravitational architecture, not diffuse mass inventory. This discrimination was not designed—the data forced it.

Scale Invariance. Rescaling the amplitude of U(z) does not change the CPL output. The shape carries the information, not the normalization. This eliminates overfitting to amplitude.

CMB Compatibility. CLASS computation with SFT's expansion history yields 0.078% RMS residual against ΛCDM across the full Planck multipole range (l = 2–2500). Acoustic peaks are untouched (< 0.2% for l > 30). The late-time ISW effect produces a ~1.8% deviation at l < 30, well within cosmic variance.

4. The Galactic Sector: Rotation Curves from the Master Equation

The master equation defines a cosmic information processing rate H_0. Multiplied by c and divided by 2π (the geometric projection from a linear expansion rate onto a spherical holographic boundary), this yields a natural acceleration threshold:

$$\mathbf{a_SFT = cH_0 / 2\pi \approx 1.05 \times 10^{-10}\ m/s^2}$$

This is the acceleration below which the cosmological solvency constraint—the same mechanism driving cosmic acceleration—modifies the local gravitational response. Compare to MOND's empirical threshold: $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. The values agree to within 15%. MOND postulated this scale in 1983 without explanation. SFT derives it from information thermodynamics.

Result: Applied to 137 quality-filtered SPARC galaxies with zero per-galaxy fitting, the weak-field law $g_{SFT}(r) = g_{bar}(r) \cdot \nu(g_{bar}/a_{SFT})$ reduces mean rotation curve error from 45.8% (baryons-only) to 16.7%—a **63% improvement** across all galaxy classes. Extended-flat galaxies improve from 36.9% to 12.7%. Short-noisy galaxies improve from 52.4% to 20.2%. Curve-intermediate galaxies improve from 44.6% to 15.1%. The improvement is broad, not class-specific.

What this does not yet show: SFT has not derived the exact interpolating function ν(x) from first principles. The tested functions are MOND-like in form. The significance is not that a MOND-like law fits SPARC—that was known—but that SFT provides a physical origin for the acceleration threshold that MOND assumes.

4.1 The Compton Maintenance Hypothesis: Origin of the 2π Factor

The postulate: particles tick at Compton frequency, each tick costs one Planck area, the master equation then produces $H^2 = 2G\rho/(3\pi)$, which differs from standard Friedmann by exactly $1/(4\pi^2)$, yielding $a_SFT = cH_0/(2\pi)$ with zero free parameters.

This explains why the SPARC data selected $cH_0/(2\pi)$ over cH_0, and identifies the dark energy gap as exactly $4\pi^2$ — an exact geometric factor rather than an unexplained empirical constant. Full derivation in the companion note SFT_V7.1_Compton_Maintenance_Hypothesis.pdf.

5. The Strong-Field Sector: Exact General Relativity

In the strong-field regime, the compactness burden $\chi = GM/(2c^2r)$ governs. The V6.1 release contains a formal symbolic proof that within the static, spherically symmetric exterior vacuum, the SFT compactness limit *structurally forces* the exact Schwarzschild metric. It does not approximate

Schwarzschild—it forbids all higher-order nonlinear deviations. When pressure is introduced, the source sector closes to the exact Tolman-Oppenheimer-Volkoff (TOV) hydrostatic equations.

This means SFT is not a modification of General Relativity. It *derives* GR's vacuum solution and its stellar interior equations from information thermodynamics. The strong-field sector is exactly GR.

6. Falsifiable Predictions

Prediction	Observable	Status
Dark energy crosses $w = -1$ near $z \approx 0.5$	DESI DR2/Y3 BAO + Pantheon+ supernovae	**Confirmed (3.4–5.4σ)**
Crossing tied to virialized halos, not raw DM	Structural specificity test on TNG300	**Confirmed (M200c works, DM fails)**
CMB compatible at background level	CLASS computation vs Planck 2018	**Confirmed (0.078% RMS)**
Galaxy acceleration threshold $\approx cH_0/2\pi$	137 SPARC rotation curves	**Confirmed (63% error reduction)**
$w(z)$ continues phantom at $z < 0.5$	DESI DR3+, Euclid, Roman	*Testable (future)*
Secular drift $\dot{G}/G \sim -H_0$	Lunar laser ranging, pulsar timing	*Testable (future)*

Dark energy gap = exactly $4\pi^2$

7. Open Problems

SFT is not a completed theory. The following gaps are acknowledged explicitly:

Calibrated parameters. The passive channel mixing ratio $\beta \approx 1.5$ and holographic scaling exponent $\mu \approx 3.5$–4.0 were swept, not derived from first principles. Physical arguments (information-weighted mass ratio, Bekenstein scaling) motivate these values but a closed derivation is needed.

Interpolating function. The galactic weak-field law uses imported MOND-like interpolation functions. Deriving $\nu(x)$ directly from the solvency constraint is an open theoretical challenge.

Perturbation theory. CMB compatibility has been confirmed at the background level only. A full SFT perturbation module in CLASS, propagating density perturbations through the SFT-modified expansion history, is needed.

Cross-simulation validation. All cosmological results use IllustrisTNG TNG300. Validation on EAGLE, SIMBA, or Magneticum would confirm independence from subgrid physics choices.

Single dynamical system. The cosmological bimodal decomposition, the galactic weak-field threshold, and the strong-field compactness burden all flow from $H = \dot{S}/I$, but they are not yet unified into a single variational principle or action formulation. This is the central theoretical task for V8.

8. Document Map

The V7 framework is modular. Each component contains a self-contained proof or result. This section maps each claim to its supporting document:

If you want to see...	Read this document
The full theoretical framework and bimodal ledger	*SFT_Academic_V6_Unified draft7.pdf*
The evolving reservoir, ablation, and scale invariance	*SFT_Note_V6_Unified.pdf*
The two-channel micro-derivation (active + passive)	*SFT_Note_V6_Companion.pdf*
Exact Schwarzschild + TOV derivation	*SFT_Local_Master_with_TOV_edit.pdf*
Galactic rotation curves (compactness burden)	*SFT Phase III Local Ledger Cost.pdf*
Weak-field SPARC test (a_SFT = $cH_0/2\pi$)	*SFT V7 Weak-Field SPARC Paper (this release)*
How SFT connects to established physics	*A Structural Synthesis of Modern Physics.pdf*
Local-to-cosmic bridge (screening + expansion)	*unified_sft_Local_Screening_Cosmic_Burden_note.pdf*
Reproducible code and data bundles	*SFT_Note_companion_verify_bundle.zip + ancillary bundle*

SFT_V7.1_Compton_Maintenance_Hypothesis.pdf

9. Summary

Solvency Field Theory proposes one equation—$H = \dot{S}/I$—and tests its consequences across five orders of magnitude in scale:

At cosmological scales, the bimodal entropy budget (active baryonic processing plus passive halo architecture) reproduces the DESI/Pantheon+ dark energy equation of state to within 0.005 in the CPL plane, with no cosmological constant. At galactic scales, the master equation yields an acceleration threshold $a_{SFT} = cH_0/2\pi$ that reduces SPARC rotation curve error by 63% with zero free parameters, providing a physical origin for MOND's empirical acceleration scale. At compact-object scales, the compactness burden structurally forces the exact Schwarzschild exterior metric and the exact TOV hydrostatic equations.

These results do not complete SFT. The interpolating function, the bimodal parameters, and the full perturbation theory remain open. But the framework demonstrates that a single information-theoretic principle, applied to physics we already have, can produce quantitative results across cosmological, galactic, and gravitational domains without new particles, new fields, or modifications to general relativity.

The full V7 framework is open access at: **10.5281/zenodo.19151148**, with Data at github.com/ShaunFosmark/Solvency-Field-Theory.